



Department of Computer Engineering
Bu-Ali Sina University
Evolutionary Computing

Evolutionary Computing 2nd Assignment

Cones Multi-Objective Optimization

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TABLE OF CONTENTS

1	Introduction	1
2	Methodology	2
2.1	Rule-base implementation	3
2.2	Python Multi-objective optimization (Pymoo) [3]	4
3	Experimental Results	5
3.1	Experiments & Results (Rule-Based & Pymoo Implementation)	5
4	Conclusion	8
	References	8

1 INTRODUCTION

Multi-objective optimization problems (MOPs) represent a class of complex challenges that involve the simultaneous optimization of multiple conflicting objectives, often leading to a set of trade-off solutions known as the Pareto front. These problems pose a unique set of difficulties, as the pursuit of improvements in one objective may result in the detriment of another. In this study, we delve into the realm of cone optimization problems, a specific subset of MOPs characterized by their geometric and mathematical structure involving cone constraints. Addressing such problems requires innovative methodologies that navigate the intricate landscape of competing objectives and constraints, ultimately aiming to uncover Pareto-optimal solutions that balance conflicting goals. Our exploration into cone optimization problems contributes to the broader understanding of multi-objective optimization and fosters the development of effective algorithms for real-world problem-solving scenarios[1].

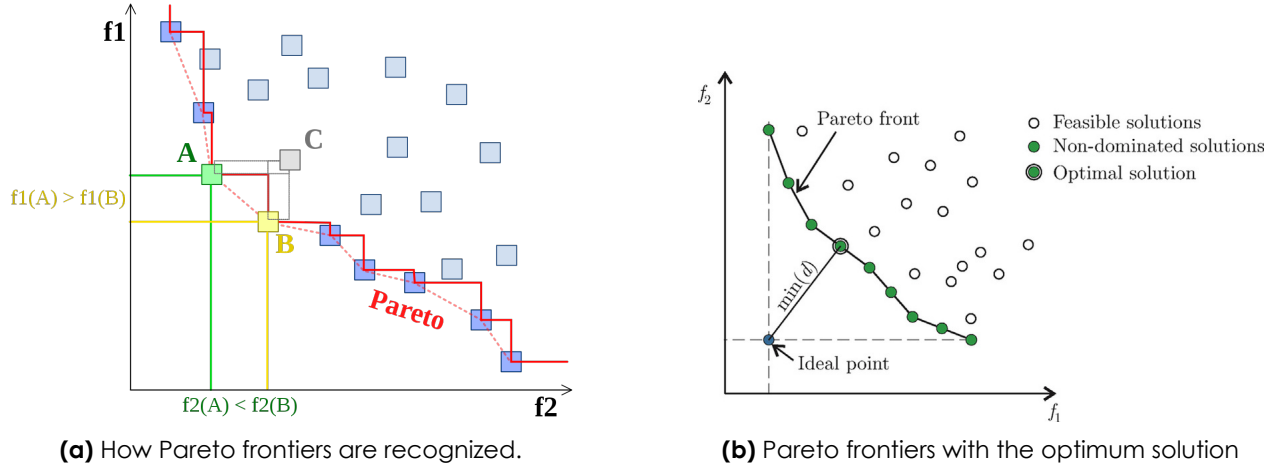


Figure 1: Pareto frontier Example.

The optimization of cone-related objectives introduces an additional layer of complexity to the multi-objective optimization landscape. Cones, being geometric structures with inherent constraints, present a unique challenge that requires specialized techniques for effective exploration and solution derivation. The study of cone optimization problems is not only pertinent to theoretical advancements in optimization algorithms but also finds practical applications in various fields such as structural engineering, signal processing, and data science. Tackling the intricacies of cone optimization contributes to the development of robust methodologies capable of handling real-world scenarios where conflicting objectives and geometric constraints coexist. Through this research, we aim to enhance the understanding of cone optimization within the broader context of multi-objective optimization, offering insights that may pave the way for improved decision-making processes and the efficient resolution of complex, real-world challenges[2].

The primary goal of multi-objective optimization (MOO) is to find a set of solutions that represents a trade-off among conflicting objectives, known as the Pareto front presented in 1. Unlike traditional single-objective optimization, where a single optimal solution is sought, multi-objective optimization recognizes that real-world problems often involve multiple, conflicting goals. MOO seeks to provide decision-makers with a range of solutions, each offering a different balance between the competing objectives. The Pareto front consists of solutions that cannot be improved in one objective without degrading another, providing a comprehensive view of the feasible solutions and the inherent trade-offs among conflicting goals.

In essence, the goal of multi-objective optimization is to provide decision-makers with a diversified and comprehensive set of solutions, enabling them to make informed choices based on their preferences and priorities. By presenting a spectrum of Pareto-optimal solutions, MOO facilitates a deeper understanding of the trade-offs inherent in complex decision-making scenarios. This approach is particularly valuable in fields such as engineering, finance, and operations research, where decisions often involve the optimization of multiple, sometimes conflicting, objectives. Ultimately, the aim is to enhance the decision-making process by offering a nuanced perspective that goes beyond the confines of single-objective optimization.

2 METHODOLOGY

In this part, we are going to investigate how to find the feasible solutions and the Pareto frontiers that exist between the feasible solutions. For educational reasons, we applied the rule-based implementation and Python multi-objective optimization module implementation.

Problem Definition: The "cones" problem is a multi-objective optimization problem that involves finding the optimal design of a right circular cone in Figure 2 by minimizing two conflicting objectives: the lateral surface area (S) and the total surface area (T). The problem has two input variables, the base radius (r) in the range of $0cm$ to $10cm$ and the height (h) in the range of $0cm$ to $20cm$, which uniquely define the shape of the cone. Additionally, there is a constraint on the volume (V) of the cone, which must be greater than 200. The Objective functions and the constraint are defined below:

- **Objective 1:** $S = \pi r s$
- **Objective 2:** $T = \pi r(r + s)$
- **Constraint:** $V > 200cm^3$

The volume(V) and the slant height(s) are calculated as follows in 1 and 2:

$$V = \frac{\pi}{3}r^2h \quad (1)$$

$$s = \sqrt{r^2 + h^2} \quad (2)$$

The challenge in this problem lies in the fact that minimizing one objective, such as the lateral surface area, may lead to an increase in the other objective, the total surface area, and vice versa. As a result, there is no single optimal solution that minimizes both objectives simultaneously. Instead, the solution is a set of designs known as the Pareto frontier, which represents the trade-off between the two objectives. The Pareto frontier consists of non-dominated solutions, where no other solution can improve one objective without degrading the other. By exploring the Pareto frontier, decision-makers can choose a solution that best balances the two objectives based on their specific requirements and priorities.

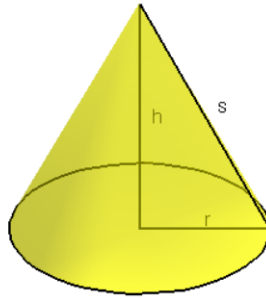


Figure 2: The cone and its parametric specifications.

2.1 Rule-base implementation

In addressing the cones optimization problem, our methodology involves the exploration of the feasible solution space to identify the Pareto-optimal frontier—a set of non-dominated solutions offering a trade-off between conflicting objectives. The problem at hand entails the simultaneous optimization of two objectives: the lateral surface area (S) and the total surface area (T) of a cone. Subject to a volume constraint, the aim is to uncover solutions that represent an optimal compromise between maximizing surface areas while adhering to the volume limitation. The identification of Pareto-optimal solutions becomes crucial in this context, as it elucidates the set of solutions that cannot be improved in one objective without compromising another.

To systematically search for solutions within the defined problem space, a grid-based approach is employed. The space is discretized by iteratively varying the radius (r) and height (h) parameters, generating a comprehensive set of candidate solutions. Each candidate solution is evaluated

based on the defined objective functions for lateral and total surface areas, as well as the volume constraint. Feasible solutions that satisfy the volume constraint are retained for further analysis and demonstrated in the "Results" sections, forming the initial set of potential Pareto-optimal solutions.

The identification of the Pareto frontier—a subset of non-dominated solutions—constitutes the final step in our methodology. Through a comparison of each feasible solution against others, those that are not dominated by any other solution with respect to both surface areas are selected. The Pareto frontier encapsulates the optimal solutions where further improvement in one objective necessitates a compromise in another. Visualization of the Pareto frontier illustrated in the next section, provides insights into the inherent trade-offs, facilitating decision-making in scenarios where multiple conflicting objectives are at play.

2.2 Python Multi-objective optimization (Pymoo) [3]

In the pursuit of solving the cones optimization problem, a Python-based multi-objective optimization library, namely PyMOO, is employed to implement the Non-Dominated Sorting Genetic Algorithm II (NSGA-II). NSGA-II is selected for its ability to efficiently explore the solution space and identify Pareto-optimal solutions, considering the inherent trade-offs between the conflicting objectives. The problem at hand involves the simultaneous optimization of two objectives: the lateral surface area (S) and the total surface area (T) of a cone. PyMOO's versatile functionalities are leveraged to define the problem, set variable bounds, and specify the constraints, offering a streamlined approach to problem formulation[4].

The ConesPB class is devised as a custom problem instance within the PyMOO framework, encapsulating the objective functions, variable bounds, and inequality constraints. This class extends the capabilities of PyMOO's Problem class, facilitating the integration of the NSGA-II algorithm for optimization. The NSGA-II algorithm is then instantiated and applied to minimize the ConesPB problem, aiming to identify Pareto-optimal solutions. The termination criteria, such as the maximum number of generations, are specified to control the algorithm's convergence.

The resulting Pareto frontiers showcase the trade-offs between the objectives, with the lateral surface area (S) plotted against the total surface area (T). This visualization aids in the interpretation of the solution set, providing insights into the range of trade-offs inherent in the cones optimization problem. Overall, the combined utilization of PyMOO and matplotlib offers a robust methodology for solving and visualizing multi-objective optimization problems, emphasizing efficiency and versatility in the exploration of complex solution spaces.

3 EXPERIMENTAL RESULTS

This section of the report will present and elucidate the outcomes of both implementations.

3.1 Experiments & Results (Rule-Based & Pymoo Implementation)

As shown in Figures 3, 4a and 4b depict the feasible solutions and Pareto frontier for the cones multi-objective optimization problem. The scatter plot shows the lateral surface area (S) on the x-axis and the total area (T) on the y-axis. The blue dots represent the feasible solutions from 100 initialized random populations that satisfy the volume constraint, while the red dots indicate the non-dominated solutions forming the Pareto frontier. In the second image, a green star marks the optimum solution with the minimum total area among the Pareto frontier solutions. The curve in Figure 4 illustrates the trade-off between the two objectives, where solutions towards the left have a smaller lateral surface area but a larger total area, and solutions towards the right have a larger lateral surface area but a smaller total area. The decision-maker can choose a solution from this Pareto front based on their preferences or priorities, considering the trade-off between the two objectives.

The 4c and 4d images focus solely on the Pareto frontier solutions, plotting the lateral surface area (S) on the x-axis and the total area (T) on the y-axis. These zoomed-in plots provide a clearer view of the trade-off between the two objectives along the Pareto frontier. The fourth image highlights the optimum solution with the minimum total area using a green star marker, allowing for easy identification of the best compromise between the two conflicting objectives based on prioritizing the minimization of the total area.

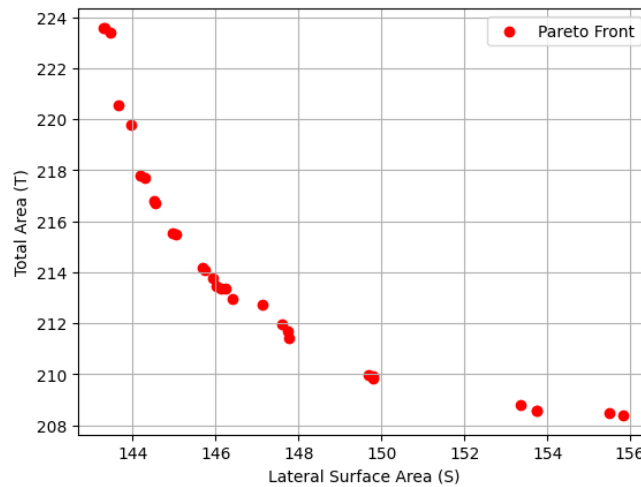
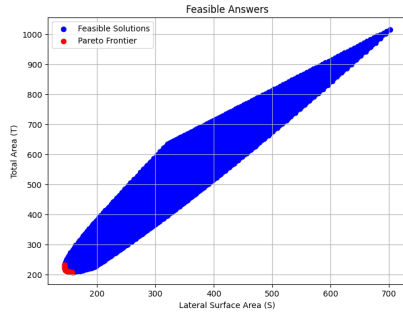
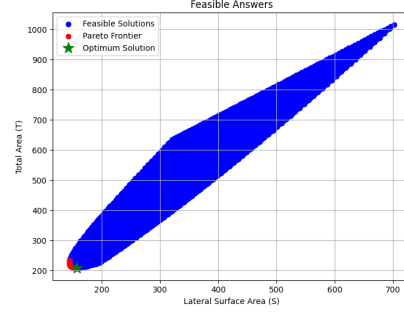


Figure 3: Find Pareto front for 100 populations using the Pymoo module.

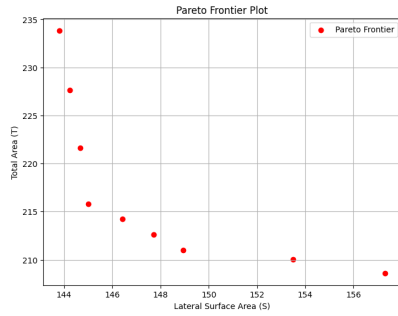
For another case study, the multi-objective optimization problem for the cones is visualized



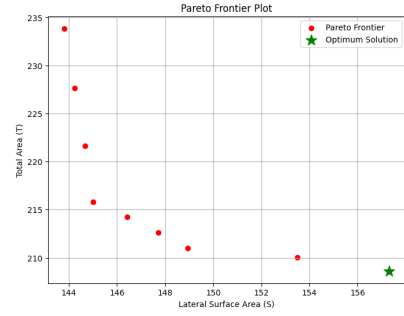
(a) Feasible answers and Pareto frontiers



(b) Feasible answers and Pareto frontiers with optimum solution.



(c) Pareto frontiers

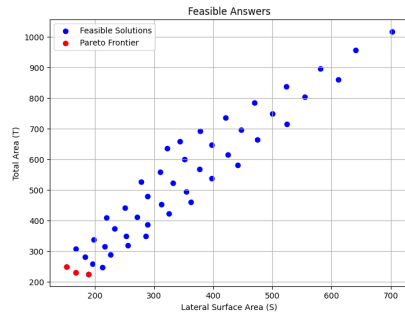


(d) Pareto frontiers and the optimum solution.

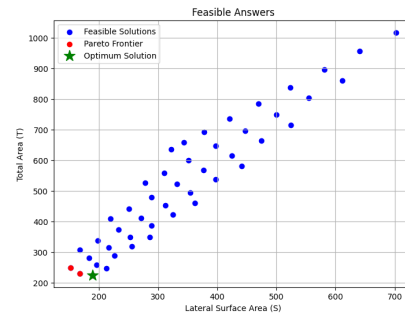
Figure 4: Illustration for 100 populations.

through scatter plots, as depicted in Figure 4. The first two figures present an overview of the feasible solutions and the Pareto frontier, which represents the non-dominated solutions that offer the best trade-off between minimizing the lateral surface area (S) and the total area (T). The feasible solutions, obtained from 10 randomly initialized populations satisfying the volume constraint, are represented by blue dots, while the red dots indicate the non-dominated solutions forming the Pareto frontier. Figure 5b further highlights the optimum solution with the minimum total area among the Pareto frontier solutions, marked by a green star.

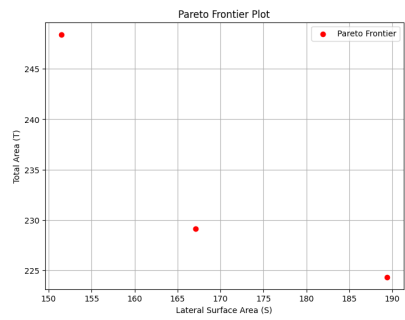
5c and 5d provide a zoomed-in perspective of the Pareto frontier, focusing solely on the lateral surface area (S) on the x-axis and the total area (T) on the y-axis. These plots offer a detailed view of the trade-off between the two objectives along the non-dominated solutions. Specifically, Figure 5d clearly identifies the optimum solution with the minimum total area, denoted by a green star marker, facilitating the selection of the best compromise when prioritizing the minimization of the total area objective.



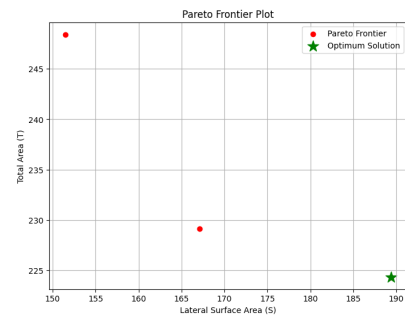
(a) Feasible answers and Pareto frontiers



(b) Feasible answers and Pareto frontiers with optimum solution.



(c) Pareto frontiers



(d) Pareto frontiers and the optimum solution.

Figure 5: Illustration for 10 populations.

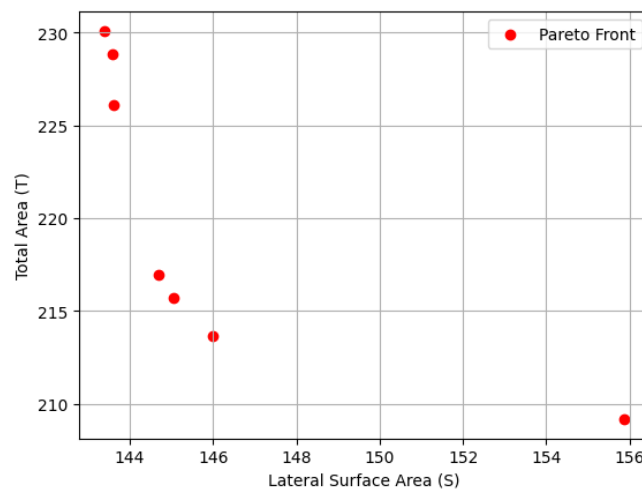


Figure 6: Find Pareto front for 10 populations using the Pymoo module.

4 CONCLUSION

In conclusion, this report has presented a comprehensive study of the cones multi-objective optimization problem, employing both a rule-based approach and the powerful PyMOO Python library. The rule-based methodology systematically explored the feasible solution space, discretizing and iterating through various combinations of cone radius and height. This rigorous process facilitated the identification of feasible solutions satisfying the volume constraint, ultimately leading to the derivation of the Pareto frontier – the set of non-dominated solutions representing optimal trade-offs between the conflicting objectives of minimizing lateral and total surface areas.

Furthermore, the utilization of PyMOO, coupled with the NSGA-II algorithm, provided a robust and efficient framework for solving the cones optimization problem. By leveraging PyMOO's versatile functionalities, the problem definition, variable bounds, and constraints were seamlessly integrated, enabling the effective exploration of the solution space. The resulting Pareto frontiers offered invaluable insights into the inherent trade-offs, aiding decision-makers in selecting solutions that best align with their priorities and preferences. The visualization capabilities further enhanced the interpretation of results, underscoring the significance of multi-objective optimization techniques in navigating complex, real-world scenarios characterized by multiple conflicting objectives.

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